01 SOURCE SYSTEMS

OLTP: ERP, CRM, POS

Flat files: CSV, Excel, logs

APIs, web data, streams/IoT

EXTRACT

Full/incremental extract; CDC and timestamps

TRANSFORM

Cleansing, validation, deduplication, conforming and standardization

LOAD

Initial / incremental; append or upsert

DATA WAREHOUSE ARCHITECTURE & DIMENSIONAL MODELING

Concept Map | Decision Support Systems (DSS)

RELATIONSHIP KEY

Solid arrows: hierarchical links | Dashed arrows: process / cross-domain links | Green: decision outcomes

02 DATA INTEGRATION: ETL/ELT

03 ARCHITECTURE

LAYERS

Source staging → storage → presentation / BI

TIERS

07 QUALITY, GOVERNANCE & SECURITY

05 OLAP & MULTIDIMENSIONAL ANALYSIS

DATA QUALITY

Accuracy Completeness. Consistency● Timeliness

GOVERNANCE

Technical/business/operational metadata Master data management Access control

Audit trails. Data lineage

08 PERFORMANCE & OPTIMIZATION

Bitmap / B-tree indexes • Partitioning Aggregate tables / materialized views Columnar storage Goal: fast, scalable and reliable analysis

DECISION VALUE

Single-tier⚫ Two-tier ⚫ Three-tier

CORE CHARACTERISTICS Subject-oriented⚫ Integrated⚫ Time-variant Non-volatile

Trusted integrated data → OLAP / BI insight

→ evidence-based decisions

→ improved planning, monitoring and forecasting

ARCHITECTURE APPROACHES

04 DIMENSIONAL MODELING

INMON CIF-top-down, normalized 3NF EDW KIMBALL BUS-bottom-up, conformed marts DATA VAULT 2.0 hubs, links, satellites HYBRID / LAKEHOUSE- combined patterns ODS operational data store

DESIGN PROCESS

1. Select business process
2. Declare the grain
3. Identify dimensions
4. Identify facts

SCHEMAS

Star⚫ Snowflake Fact constellation / Galaxy

ROLAP
OLAP ENGINES

relational + star schema MOLAP - multidimensional array
HOLAP - hybrid

CUBE ELEMENTS Dimensions Measures. Hierarchy levels Aggregations/lattice of cuboids

OPERATIONS

Roll-up Drill-down Slice Dice ⚫ Pivot
06 DECISION SUPPORT & BI

USERS & DECISION LEVELS Executives - strategic planning Business analysts - tactical decisions Managers — operational monitoring

BI TOOLS Dashboards/KPIs⚫ Ad-hoc queries Reports Data mining / ML⚫ Visualization
FACT TABLES & MEASURES
DIMENSION TABLES & HISTORY

FACT TYPES Transaction-one row per event Periodic snapshot — one row per period Accumulating snapshot-process lifecycle Factless fact-events/coverage

MEASURE BEHAVIOR Additive⚫ Semi-additive⚫ Non-additive
DIMENSION ROLES Conformed⚫ Role-playing⚫ Junk ⚫ Degenerate Surrogate keys identify dimension rows Hierarchies: Day → Month → Year

SCD TYPES

Type O retain original⚫ Type 1 overwrite Type 2 add new row⚫ Type 3 add new column Type 6 hybrid (Type 1 + 2 + 3)